

46 YEARS OF EXCELLENCE | SRAMAYEVA JAYATHE | NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR - A & APEX, ALL IITS **DT: 22/01/2026**
Time: 3 Hrs **GTM - 22** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. Two harmonic waves moving in the same direction superimpose to form a wave $x = a\cos\left(\frac{\pi}{4}t\right)\cos\left(20\pi t\right)$ where t is in seconds. Find the beat period

A) 6s B) 4s C)1s D) 2s

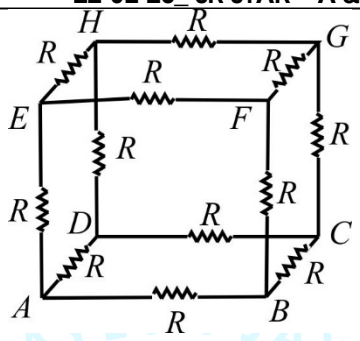
2. Two plane polarized light waves combine at a point. Electric field components of the individual waves are $E_1 = E_0 \sin\left(\omega t + \frac{\pi}{3}\right)$, $E_2 = 2E_0 \sin\left(\omega t - \frac{\pi}{3}\right)$

Electric field amplitude of the resultant wave is

A) $E_0\sqrt{3}$ B) $3E_0$ C) $E_0\sqrt{7}$ D) $E_0/\sqrt{2}$

3. 12 identical resistors, each of resistance R are connected, each being an edge of a cube as shown in the figure

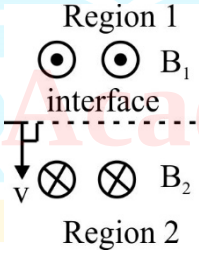
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What is the equivalent resistance between the two adjacent terminals A and B?

A) $5R/6$ B) $6R/5$ C) $12R/7$ D) $7R/12$

4. Uniform magnetic fields of different strength (B_1 and B_2), both normal to the plane of the paper exist as shown in the figure. A charged particle of mass m and charge q, at the interface at an instant, moves into the region 2 with velocity v and returns to the interface. It continues to move into region 1 and finally reaches the interface. What is the displacement of the particle during this movement along the interface?



(Consider the velocity of the particle to be normal to the magnetic field and $B_2 > B_1$)

A) $\frac{mv}{qB_1}\left(1-\frac{B_2}{B_1}\right)\times 2$ B) $\frac{mv}{qB_1}\left(1+\frac{B_1}{B_2}\right)$

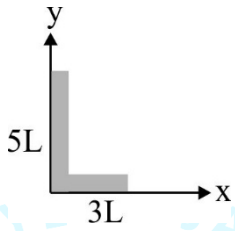
C) $\frac{mv}{qB_1}\left(1-\frac{B_2}{B_1}\right)$ D) $\frac{mv}{qB_1}\left(1+\frac{B_1}{B_2}\right)\times 2$

5. Assume that energy E, angular momentum J, mass M and universal gravitational constant G are the fundamental quantities in a system, then the dimension of $\frac{EJ^2}{M^5G^2}$ is similar to

A) Energy B) Momentum C) Power D) Angle

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6. A thin, uniform rod of length 8L is bent at right angle keeping one side length as 3L



The position of the centre of mass of the system:

A) $L\left(3\hat{i}+5\hat{j}\right)$ B) $\frac{L}{16}\left(9\hat{i}+25\hat{j}\right)$ C) $\frac{L}{8}\left(3\hat{i}+5\hat{j}\right)$ D) $\frac{L}{5}\left(6\hat{i}+10\hat{j}\right)$

7. A long solenoid has 1500 turns per metre and an iron core $\mu_r = 1100$ is kept inside it. 2 A current flows in the coil of solenoid. If core is heated beyond Curie temperature then

A) The H and B fields in the solenoid reduce to zero

B) The H and B fields in the solenoid remain unchanged

C) The H field in the solenoid is nearly unchanged but B field decreases significantly

D) The B field in the solenoid is nearly unchanged but H field decreases significantly

8. An equiconvex glass lens has focal length 40 cm in air. This lens is cut into two identical Plano-convex lenses. One of them is immersed in water. What is its focal length in water? ($\mu_{glass} = \frac{3}{2}$ and $\mu_{water} = \frac{4}{3}$)

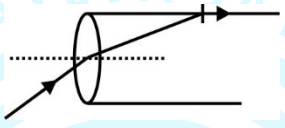
A) 320 cm B) 80 cm C) 360 cm D) 120 cm

9. An alternating voltage is given by $V = 4 + 7\cos\left(850\pi t + \frac{\pi}{8}\right)$ Volt. The r.m.s value of this voltage is ___Volt

A) 18 B) $9\sqrt{2}$ C) $9/\sqrt{2}$ D) $\sqrt{22}$

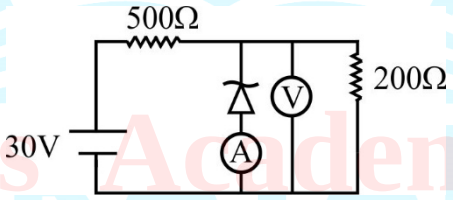
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10. A cylindrical solid rod made up of material of refractive index $\frac{2}{\sqrt{3}}$ is given. A light beam enters from one circular end of the cylinder, and comes out just brushing the outer surface along the length. What should be the angle of incidence on the circular face?



A) $\sin^{-1}\left(\frac{1}{2}\right)$ B) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$ C) $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$ D) $\sin^{-1}\left(\frac{2}{\sqrt{3}}\right)$

11. In the following circuit, the reading of the ammeter will be(Take Zener breakdown voltage=4V)



A) 30 mA B) 32 mA C) 20 mA D) 40 mA

12. Two particles are thrown from ground with same speed for same horizontal range R on different paths. If T_1 and T_2 are their time of flight, then T_1T_2 is

A) $\frac{R}{g}$ B) $\frac{2R}{g}$ C) $\frac{R^2}{g^2}$ D) $\frac{4R^2}{g^2}$

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13.

List-I		List-II	
A.	Triatomic rigid gas	I.	$f = 3$
B.	Diatomic non-rigid gas	II.	$C_V = \frac{5R}{2}$
C.	Monatomic gas	III.	$\frac{C_P}{C_V} = \frac{4}{3}$
D.	Diatomic rigid gas	IV	$C_P = \frac{9R}{2}$

Choose the correct answer from the option given below:

A) A-IV,B-II,C-I,D-III

B) A-III, B-I, C-IV, D-II

C) A-III, B-IV, C-I, D-II

D) A-II, B-I, C- IV, D-III

14. A block is projected up a rough incline making an angle 30^0 with the horizontal. If the time of descent is 40% more than the time of ascent, then the co-efficient of friction between the block and the incline is

A) $\frac{5}{9\sqrt{3}}$ B) $\frac{12}{25\sqrt{3}}$ C) $\frac{16}{35}$ D) $\frac{4\sqrt{3}}{37}$

15. An excited H-atom emits a photon of wavelength λ in returning to the ground state. The quantum number n of excited state is given by (R = Rydberg constant)

A) $\sqrt{\lambda R(\lambda R - 1)}$

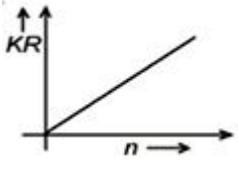
B) $\sqrt{\frac{\lambda R}{(\lambda R - 1)}}$

C) $\sqrt{\frac{\lambda R - 1}{\lambda R}}$

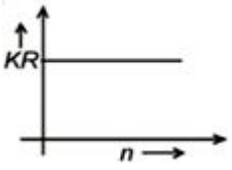
D) $\sqrt{\frac{1}{\lambda R(\lambda R - 1)}}$

16. If R represents the orbital radius of an electron moving in an orbit and K represents its kinetic energy. Then the quantity KR varies with principal quantum number n as

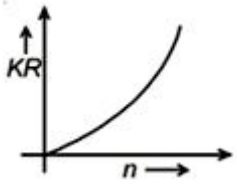
A)

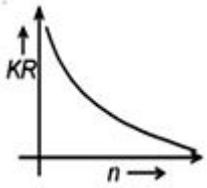


B)

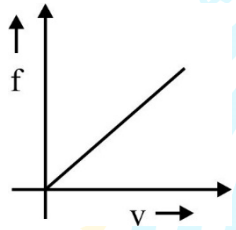


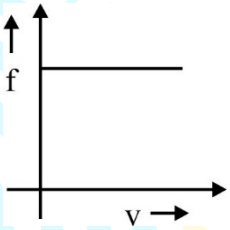
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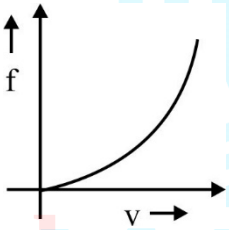
C) 

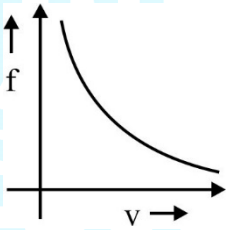
D) 

17. A particle of charge q mass m is projected at speed v perpendicular to a uniform magnetic field B. which of the following graphs represents the variation of frequency of rotation f with v

A) 

B) 

C) 

D) 

18. An object of mass 500g starts from rest and moves under the influence of a force $\vec{F} = 4t^3\hat{i} - 3t^2\hat{j} - \hat{k}$. Where t is time in seconds. The power due to this force as a function of time is

A) $2\left[4t^7 + 3t^5 + t\right]$ B) $2\left[4t^7 - 3t^5 + 2t\right]$

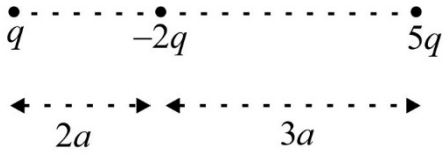
C) $\left[8t^7 - 6t^5 - t\right]$ D) $2\left[4t^7 - 3t^5 - t\right]$

19. Two wires A and B are made of same material having ratio of lengths $\frac{L_A}{L_B} = \frac{4}{3}$ and their diameters ratio $\frac{d_A}{d_B} = \frac{2}{5}$. If both the wires are stretched using same force, what would be the ratio of their respective elongations?

A) 3:25 B) 16:75 C) 75:12 D) 25:3

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20. 3 point charges $q, -2q$ and $5q$ are kept on a straight line as shown in the figure.



Find the work done in rearranging the system into an equilateral triangle of side a

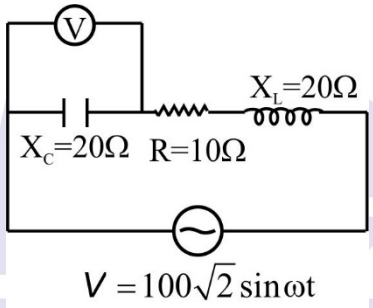
A) $\frac{kq^2}{3a}$ B) $\frac{-11kq^2}{5a}$ C) $\frac{-11kq^2}{3a}$ D) $\frac{23kq^2}{3a}$

SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

21. 3 identical thin uniform rods each of mass m and length l are joined to form an equilateral triangle. Moment of inertia of this system about an axis through the mid-point of one rod, perpendicular to the plane of the triangle is $\frac{x}{4}ml^2$. Then x is____

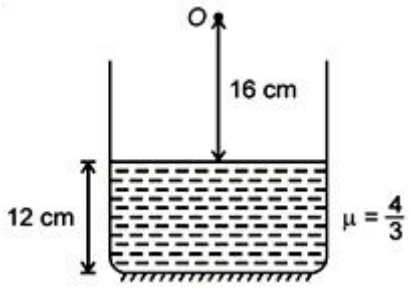
22.



Reading of the AC voltmeter in the above figure is x Volt. Then $\frac{x}{100}$ is ___

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23.



Bottom of the container is silvered. The distance between the object and its final image is__ cm

24. Temperature of the surface of the sun is given by $T=\left[\frac{s}{\sigma}\left(\frac{R}{r}\right)^{a}\right]^{b}$ where

s is solar constant (energy from sun received per unit area per unit time on earth)

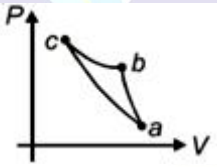
σ is Stefan's constant

r is distance between sun and earth

R is radius of the sun

Then $a+\frac{1}{b}$ is ___

25. In the given pressure-volume (P-V) diagram, the gas does 5J of work along isotherm cb and 4J along adiabatic ba. What is the change in the internal energy of the gas if the gas traverses the straight path from a to c(in Joule)



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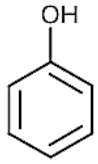
CHEMISTRY **MAX.MARKS: 100**

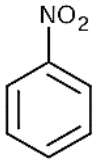
SECTION – I
(SINGLE CORRECT ANSWER TYPE)

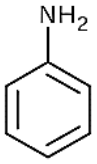
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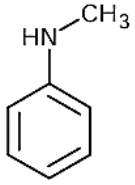
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. Which of the following gives Libermann nitroso reaction?

A) 

B) 

C) 

D) 

A) A and B B) B,C and D C) A and D D) Only D

27. Match the following List-I and List-II

List-I (Molecule)		List-II (Number of d-orbital(s) used in bond formation)	
(P)	SOF_4	(1)	1
(Q)	XeO_3F_2	(2)	2
(R)	XeO_3	(3)	3
(S)	$POCl_3$	(4)	4

A) P-4, Q-1, R-2, S-3 B) P-2, Q-4, R-3, S-1

C) P-2, Q-1, R-3, S-4 D) P-3, Q-4, R-2, S-1

28. Statement-I : Hydrolysis of sucrose brings a change in sign of optical rotation

Statement-II: Specific rotation of fructose is -92.4^o and Glucose has $+52.5^o$

Choose the correct option

A) Both Statement I and Statement II are correct

B) Both Statement I and Statement II are incorrect

C) Statement I is correct and Statement II is incorrect

D) Statement I is incorrect and Statement II is correct

29. Correct order of frequency of the following radiations is

I-Gamma rays II-Infrared rays III-UV rays IV-Micro waves

A) I>II>III>IV B) I>III>II>IV

C) I<II<III<IV D) I<III<II<IV

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30. Statement I: The conductivity of solutions of different electrolytes in the same solvent and at a given temperature is same.

Statement II: The conductivity depends on the charge and size of the ions, the concentrations of ions and ease with which the ions move under potential gradient.

Choose the correct option

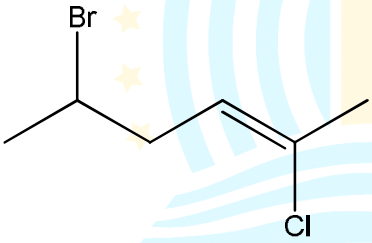
A) Both statement I and II are correct

B) Both statement I and II are incorrect

C) Statement I is correct and Statement II is incorrect

D) Statement I is incorrect and Statement II is correct

31. IUPAC name of the given compound is



A) 2 – Chloro – 5 – bromo – hex-2-ene

B) 5 – Bromo – 2 – chloro-hex-2-ene

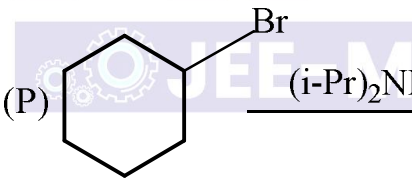
C) 2 – Bromo – 5 – chloro hex-4-ene

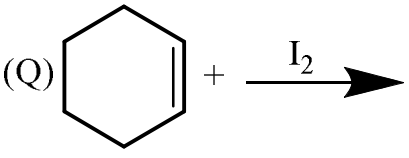
D) 5 – Chloro – 2 – chloro hex-4-ene

32. A metal M has highest enthalpy of atomization in 3d series. The maximum oxidation number exhibited by the metal M is

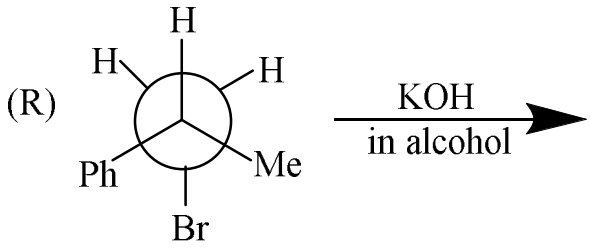
A) +6 B) +5 C) +4 D) +3

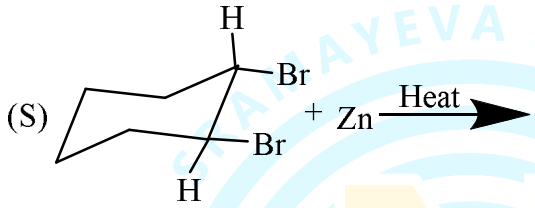
33. The reactions from which alkene will not be produced among the following is





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A) P and R B) Q and R C) Only Q

D) Alkene is produced in every reaction

34. What is the pH of solution made by mixing equal volumes of 0.1N H_2SO_4, 0.1N HNO_3 and 0.1N HCl?

A) 1 B) 2 C) 3 D) 4

35. The speed of electron revolving in the fourth orbit of a hydrogen-like atom or ion is 1635 km/s. The atom or ion is

A) H B) He^+ C) Li^{2+} D) Be^{3+}

36. The enthalpy of neutralization of a weak monoprotic acid (HA) in $1M$ solution with a strong base is $-55.95 kJ/mol$. If the unionized acid required $1.4 kJ/mol$ heat for its complete ionization and enthalpy of neutralization of the strong monobasic acid with a strong monoacidic base is $-57.3 kJ/mol$. The $\left[H^+\right]$ in the given acid solution is____

A) 1.0 M B) 0.0357 M C) 1.35 M D) 0.02 M

37. X and Y from the following equation are

$[X] + H_2SO_4(dilute) \rightarrow [Y]$, a colourless gas

$[Y] + K_2Cr_2O_7 + H_2SO_4 \rightarrow$ green solution

A) SO_3^{2-}, SO_2 B) Cl^-, HCl C) SO_4^{2-}, H_2S D) CO_3^{2-}, CO_2

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38. Statement I: Ozonolysis of 1,2 dimethyl benzene gives three different products

Statement II: All three products contain atleast one sp^3 carbon

Choose the correct option

A) Both statement I and II are correct

B) Both statement I and II are incorrect

C) Statement I is correct and Statement II is incorrect

D) Statement I is incorrect and Statement II is correct

39. Statement 1: $CH_2 = CH - OH$ more acidic than $CH_3 - CH_2 - OH$

Statement 2: Enolate ion is stabilised by resonance

A) STATEMENT–1 is True, STATEMENT–2 is True; STATEMENT–2 is a Correct explanation for STATEMENT–1

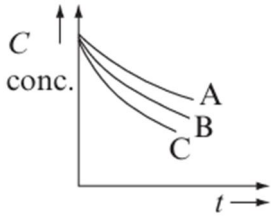
B) STATEMENT–1 is True, STATEMENT–2 is True; STATEMENT–2 is NOT a Correct explanation for STATEMENT–1

C) STATEMENT–1 is True, STATEMENT–2 is False

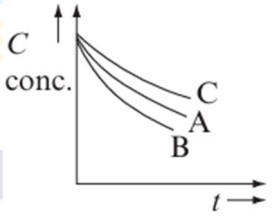
D) STATEMENT–1 is False, STATEMENT–2 is True

40. Which of the following graphs is correct representation for the reaction of type $A + 2B \rightarrow C$? (assume initial concentration of A and B is same)

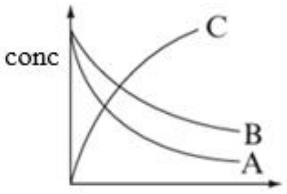
A)



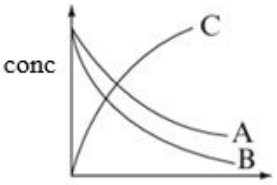
B)



C)



D)



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41. One litre of mixture of oxygen and ozone at STP (1.0 atm, 273K) was allowed to react with excess of acidified solution of KI. Released iodine required 80 mL of M/20 sodium thiosulphate solution for titration. Volume of the oxygen in original mixture at STP will be

$$2KI + H_2O + O_3 \rightarrow 2KOH + I_2 + O_2$$

$$I_2 + 2Na_2S_2O_3 \rightarrow Na_2S_4O_6 + 2NaI$$

A) 955.2 mL B) 800 mL C) 0.056 L D) 955.2 L

42. $\left[CoCl_2\left(NH_3\right)_4\right]^+ + Cl^- \rightarrow \left[CoCl_3\left(NH_3\right)_3\right] + NH_3$

If in this reaction two isomers of the product are obtained, which is true for the initial (reactant) complex:

A) Compound is in cis-forms

B) Compound is in trans-form

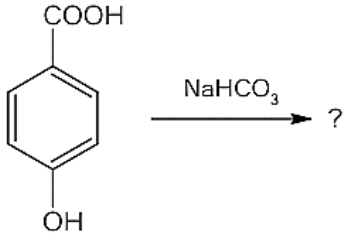
C) Compound is in both (cis and trans) forms

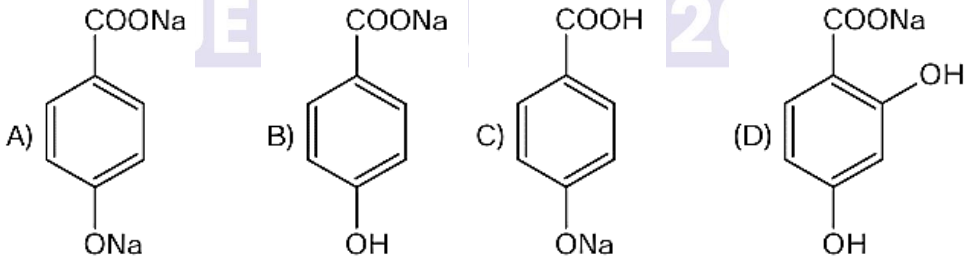
D) Cannot be predicted

43. Which of the following metals cannot be detected in the laboratory through flame test

A) Na B) Mg C) K D) Ba

44. The product of the following reaction is

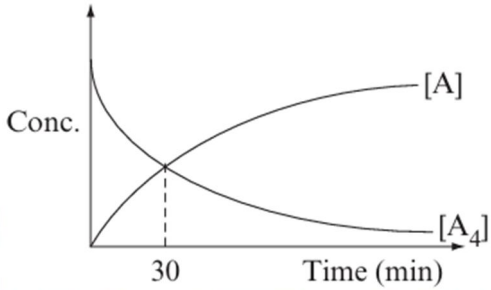




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45. Consider the following first-order decomposition reaction. $A_4(g) \rightarrow 4A(g)$

Which of the following statement(s) is incorrect regarding the reaction? $(log 2 = 0.3)$



A) At 30 min, only 20% reaction is complete.

B) $t_{1/2}$ of reaction is nearly 90 min.

C) Rate of reaction decreases linearly with time.

D) The time for intersection of two curves is independent of initial concentration of A_4

SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

46. On complete combustion, 0.246 g of an organic compound gave 0.198g of carbon dioxide and 0.1014g of water. Determine the percentage composition of carbon in the compound (nearest integer)

47. How many of the following complexes can exist as 'd' and 'l' isomers

$[Co(NH_3)_4Cl_2]^+, [Co(NH_3)_3Cl_3], trans[Co(en)_2Cl_2]^+,$

$Cis[Co(en)_2Cl_2]^+, [Cr(C_2O_4)_3]^{3-}, [Cr(en)_3]^{3+}, Cis[Pt(en)_2ClBr]$

48. Aqueous solutions of 0.004 $M - Na_2SO_4$ and 0.01 M-Glucose are isotonic. The percentage dissociation of Na_2SO_4 is

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49. The EMF of a galvanic cell with following cell reaction

$$2Cr(s)+3Cu^{2+}_{(aq)} \rightarrow 2Cr^{3+}_{(aq)}+3Cr_{(s)};E^0=0.43$$

is measured as $P\times10^{-2}V$.

Given that $\left[Cu^{2+}\right]=1.0M$ and $\left[Cr^{3+}\right]=0.01M$

The value of P is____

50. A tranquilizer Equanil (Mol.Wt 190) has 33.68% oxygen. Number of oxygen atoms per molecule of the tranquiliser is

MATHEMATICS MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. Each of the $'n'$ urns contain 4 white and 6 black balls. The $(n+1)^{th}$ urn contains 5 white and 5 black balls. One of the urn is chosen at random and two balls are drawn from it without replacement. Both the balls turn out to be black. If the probability that the $(n+1)^{th}$ urn was chosen to draw the balls is $\frac{1}{16}$, then the value of n is

A) 10 B) 11 C) 12 D) 13

52. The area cut off the parabola $4y=3x^2$ by the straight line $2y=3x+12$ in sq. units is

A) 27 B) 2π C) $\frac{32}{3}$ D) $\frac{11}{4}$

53. Shortest distance between the lines $\frac{x-1}{1}=\frac{y-1}{1}=\frac{z-1}{1}$ and $\frac{x-2}{1}=\frac{y-3}{1}=\frac{z-4}{1}$ is equal to

A) $\sqrt{14}$ B) $\sqrt{7}$ C) $\sqrt{2}$ D) None of these

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54. If $f(x)=a\ln|x|+bx^2+x$ has extremas at $x=1$ and $x=3$, then

A) $a=\frac{3}{4},b=-\frac{1}{8}$ B) $a=\frac{3}{4},b=\frac{1}{8}$

C) $a=-\frac{3}{4},b=-\frac{1}{8}$ D) $a=-\frac{3}{4},b=\frac{1}{8}$

55. $\vec{a}$ and $\vec{c}$ are unit vectors and $|\vec{b}|=4$. The angle between $\vec{a}$ and $\vec{c}$ is $\cos^{-1}\left(\frac{1}{4}\right)$ and $\vec{b}-2\vec{c}=\lambda\vec{a}$. The value of λ is

A) $3,-4$ B) $\frac{1}{4},\frac{3}{4}$ C) $-3,4$ D) $\frac{-1}{4},\frac{3}{4}$

56. In a G.P. the first, third, and fifth terms may be considered as the first, fourth, and sixteenth terms of an A.P. then the fourth term of the A.P. knowing that its first term is 5, is

A) 10 B) 12 C) 16 D) 20

57. All the values of m for which both the roots of the equation $x^2-2mx+m^2-1=0$ are greater than -2 but less than 4 lie in the interval

A) $-2<m<0$ B) $m>3$ C) $-1<m<3$ D) $1<m<4$

58. The fractional part of $\frac{32^{32}}{3}$ is equal to

A) 0 B) $\frac{1}{3}$ C) $\frac{1}{5}$ D) $\frac{1}{6}$

59. The value of $\lim\limits_{x\to 0}\left((\cos x)^{\frac{1}{\sin^2 x}}+\frac{\sin 2x+2\tan^{-1}3x+3x^2}{\ln(1+3x+\sin^2 x)+xe^x}\right)$ is:

A) $\sqrt{e}+\frac{3}{2}$ B) $\frac{1}{\sqrt{e}}+\frac{3}{2}$ C) $\sqrt{e}+2$ D) $\frac{1}{\sqrt{e}}+2$

60. A function $y=f(x)$ satisfies the differential equation $(x+1)f'(x)-2(x^2+x)f(x)=\frac{e^{x^2}}{(x+1)}$; $\forall x>-1$. If $f(0)=5$, then $f(x)$ is :

A) $\left(\frac{3x+5}{x+1}\right).e^{x^2}$ B) $\left(\frac{6x+5}{x+1}\right).e^{x^2}$ C) $\left(\frac{6x+5}{(x+1)^2}\right).e^{x^2}$ D) $\left(\frac{5-6x}{x+1}\right).e^{x^2}$

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61. The number of ways of choosing a committee of 2 women &3 men from 5 women &6 men. If Mr. A refuses to serve on the committee if Mr. B is a member &Mr. B can only serve if Miss C is the member of the committee, is

A) 60 B) 84 C) 124 d) 120

62. $K_1 < K_2$ be the values of K for which the system of linear equations $Kx+4y+z=0, 4x+Ky+2z=0$ and $2x+2y+z=0$ has non-trivial solution. Let $P=\begin{bmatrix} K_2 & K_1 \\ K_1 & K_2 \end{bmatrix}$. Then $P^{-1}=$

A) $\frac{1}{5}\begin{bmatrix} 3 & -2 \\ 2 & 3 \end{bmatrix}$ B) $\frac{1}{12}\begin{bmatrix} 4 & -2 \\ -2 & 4 \end{bmatrix}$ C) $\frac{1}{5}\begin{bmatrix} 3 & -2 \\ -2 & 3 \end{bmatrix}$ D) $\frac{1}{7}\begin{bmatrix} 4 & -3 \\ -3 & 4 \end{bmatrix}$

63. If A is a square matrix of order 3 such that $|A|=2$ then $\left|\left(adjA^{-1}\right)^{-1}\right|$ is

A) 1 B) 2 C) 4 D) 8

64. The value of $\int_0^{\pi/2} \frac{x\sin x\cos x}{\cos^4 x+\sin^4 x}dx$ is

A) $\frac{\pi^2}{2}$ B) $\frac{\pi^2}{4}$ C) $\frac{\pi^2}{8}$ D) $\frac{\pi^2}{16}$

65. The axis of a parabola is along the line $y=x$ and the distance of its vertex and focus from origin are $\sqrt{2}$ and $2\sqrt{2}$ respectively. If vertex and focus both lie in the first quadrant, then the equation of the parabola is

A) $(x+y)^2=(x-y-2)$ B) $(x-y)^2=(x+y-2)$

C) $(x-y)^2=4(x+y-2)$ D) $(x-y)^2=8(x+y-2)$

66. In the interior of an equilateral triangle, 3 coins of radii '1' are kept so that they touch each other and also the sides of the triangle, then the Area of the triangle is

A) $4+2\sqrt{3}$ B) $6+4\sqrt{3}$ C) $12+\frac{7\sqrt{3}}{4}$ D) $3+\frac{7\sqrt{3}}{4}$

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67. Two vertices of a triangle are $(4,-3)$ and $(-2,5)$. Orthocentre of this triangle is $(1,2)$. Then the coordinates of its third vertex is

A) $(30,20)$ B) $(33,26)$ C) $(35,28)$ D) $(35,26)$

68. The mean and standard deviation of 20 observations are found to be 10 and 2, respectively, it was found that an observation by mistake was taken 8 instead of 12. The correct standard deviation is

A) $\sqrt{3.86}$ B) 1.8 C) $\sqrt{3.96}$ D) 1.94

69. Given $\frac{x}{a}+\frac{y}{b}=1$ and $ax+by=1$ are two variable lines, 'a' and 'b' being the parameters connected by the relation $a^2+b^2=ab$. The locus of the point of intersection has the equation

A) $x^2+y^2+xy-1=0$ B) $x^2+y^2-xy+1=0$

C) $x^2+y^2+xy+1=0$ D) $x^2+y^2-xy-1=0$

70. Given z is a complex number with modulus 1. Then the equation in a, $\left[\frac{(1+ia)}{(1-ia)}\right]^4=z$ has

A) All roots real and distinct B) Two real and two imaginary

C) Three roots real and one imaginary D) one root real and three imaginary

SECTION – II
(Numerical Value Answer Type)

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71. If 3×3 matrices are formed using the elements $-1,0,1$ then the number of matrices having trace equal to 2 is

72. The number of points $f(x)=\begin{cases}[\cos\pi x], & 0\le x\le 1\\ |2x-3|[x-2], & 1<x\le 2\end{cases}$ is discontinuous is___

([.] denotes the greatest integral function)

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73. Let $A=\{1,2,3,4\}$ and $R=\{(1,2),(2,3),(1,4)\}$ be a relation on A. Let S be the equivalence relation on A such that $R\subset S$ and the number of elements in S is n. then the minimum value of n is

74. Let $S=\{1,2,3,4,5,6,7\}$. Then the number of possible functions $f:S\to S$ such that $f(m.n)=f(m).f(n)$ for every $m,n\in S$ and $m.n\in S$ is equal to___

75. Let the latus rectum of the hyperbola $\frac{x^2}{9}-\frac{y^2}{b^2}=1$ subtend an angle of $\frac{\pi}{3}$ at the centre of the hyperbola. If b^2 is equal to $\frac{l}{m}\left(1+\sqrt{n}\right)$, where l and m are co-prime numbers and n is integer, then $l^2+m^2+n^2$ is equal to___

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